Polar Equations and Graphs

Section 1: Circles and Lines in Polar Form

1 What is the graph of $r = a$ where a is a constant?

a. A circle centered at the origin with radius $|a|$

b. A line through the origin

c. A parabola

d. An ellipse

2 What is the graph of $\theta = \alpha$ where α is a constant angle?

a. A straight line passing through the origin with slope $\tan\alpha$

b. A circle

c. A ray only

d. A spiral

3 Describe the circle given by $r = -6\sin\theta$.

a. Centered at $(0, -3)$ with radius 3

b. Centered at $(0, 3)$ with radius 3

c. Centered at $(0, 0)$ with radius 6

d. Centered at $(-3, 0)$ with radius 3

Section 2: Limaçons and Cardiods

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4 Under what condition does the limaçon $r = a + b\cos\theta$ $(a, b > 0)$ form a cardioid (heart-shaped curve)?

a. $a/b = 1$ (i.e. $a = b$)

b. $a/b < 1$

c. $1 < a/b < 2$

d. $a/b \geq 2$

5 Which equation produces a limaçon with an inner loop?

a. $r = 2 + 3\cos\theta$

b. $r = 3 + 3\cos\theta$

c. $r = 4 + 2\cos\theta$

d. $r = 3 + 2\cos\theta$

6 What type of curve is $r = 5 + 2\sin\theta$?

a. A convex (dimple-free) limaçon

b. A cardioid

c. A limaçon with an inner loop

d. A dimpled limaçon

Section 3: Rose Curves

7 How many petals does the rose curve $r = 4\cos(3\theta)$ have?

a. 3 petals

b. 6 petals

c. 4 petals

d. 12 petals

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8 How many petals does the rose curve $r = 2\sin(4\theta)$ have?

a. 8 petals

b. 4 petals

c. 16 petals

d. 2 petals

9 What is the maximum length of each petal in $r = 5\cos(2\theta)$?

a. 5

b. 10

c. 2.5

d. 2

Section 4: Lemniscates and Spirals

10 What figure-eight-shaped curve is given by $r^2 = 9\cos(2\theta)$?

a. A lemniscate

b. A cardioid

c. A rose curve

d. A spiral of Archimedes

11 What type of curve is represented by the polar equation $r = 2\theta$ for $\theta \geq 0$?

a. Spiral of Archimedes

b. Logarithmic spiral

c. Lemniscate

d. Circle

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12 Which symmetry test verifies that a polar graph is symmetric with respect to the polar axis (the horizontal axis)?

a. Replacing (r, θ) with $(r, -\theta)$ yields an equivalent equation

b. Replacing (r, θ) with $(-r, \theta)$

c. Replacing θ with $\theta + \pi$

d. Replacing r with $1/r$

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